

1. Cloud Computing (Basics)

Definition

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, and software over the Internet instead of using your own computer or local server.

Example

Instead of saving files only on your laptop, you save them in **Google Drive**. You can access them from any device connected to the Internet.

Why Cloud Computing?

- Access data from anywhere.
- Reduce hardware costs.
- Increase or decrease resources as needed (scalability).
- Automatic backup and recovery.
- High availability and reliability.

Examples of Cloud Services

- **AWS (Amazon Web Services)**
 - **Microsoft Azure**
 - **Google Cloud Platform (GCP)**
-

2. Absolute Path

Definition

An **Absolute Path** is the complete path to a file or directory starting from the **root directory (/)**. It always points to the same location, no matter where you are currently in the file system.

Syntax

/root/folder/subfolder/file

Example

Suppose a file is located at:

/home/student/Documents/report.txt

You can access it from **any directory** using:

```
cat /home/student/Documents/report.txt
```

Advantages

- Gives the exact location of a file.
 - Works from any directory.
 - Easy to identify where the file is stored.
-

3. Relative Path

Definition

A **Relative Path** specifies the location of a file **relative to the current working directory**. It does not start with /.

Example

Current directory:

/home/student

To access:

/home/student/Documents/report.txt

Use:

```
cat Documents/report.txt
```

If you move to another directory, this relative path may no longer work because it depends on your current location.

Advantages

- Shorter and easier to type.
 - Useful when working within the same project.
-

Difference Between Absolute Path and Relative Path

Absolute Path

Starts from the root directory (/)

Always begins with /

Independent of current location

Example: /home/student/file.txt

Relative Path

Starts from the current directory

Does not begin with /

Depends on current location

Example: Documents/file.txt

4. Linux Basic Commands

1. pwd (Print Working Directory)

Displays the current directory.

pwd

Example Output:

/home/student

2. ls (List)

Lists all files and directories in the current location.

ls

3. ls -l

Displays files with detailed information such as permissions, owner, size, and modification date.

ls -l

4. ls -a

Shows all files, including hidden files (those beginning with .).

ls -a

5. cd (Change Directory)

Moves to another directory.

cd Documents

Move to the parent directory:

cd ..

Move to the home directory:

cd ~

6. mkdir (Make Directory)

Creates a new directory.

mkdir Project

7. rmdir

Deletes an empty directory.

rmdir Project

8. touch

Creates a new empty file.

touch file.txt

9. cat

Displays the contents of a file.

cat file.txt

10. cp (Copy)

Copies a file from one location to another.

```
cp file1.txt file2.txt
```

11. mv (Move/Rename)

Moves a file or renames it.

Rename:

```
mv old.txt new.txt
```

Move:

```
mv file.txt Documents/
```

12. rm (Remove)

Deletes a file.

```
rm file.txt
```

Delete a directory and its contents:

```
rm -r foldername
```

13. clear

Clears the terminal screen.

```
clear
```

14. whoami

Displays the name of the current logged-in user.

```
whoami
```

15. date

Displays the current date and time.

```
date
```

16. cal

Displays the current month's calendar.

cal

- **Cloud Computing:** Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the Internet on demand.
- **Absolute Path:** The complete path from the root directory (/) to a file or directory.
- **Relative Path:** The path to a file or directory relative to the current working directory

On-Demand Self-Service: Users do not have to communicate with the IT staff of a service provider in order to provision and configure computing resources (servers, storage, databases, etc.). The self-service aspect makes it simple and quick to scale resources as needed.

Broad Network Access: Cloud services can be accessed via the internet using a variety of devices (laptops, desktops, tablets, and smartphones) from any location in the globe. This makes remote work and teamwork possible.

Resource Pooling: To service several clients, cloud companies combine their physical and virtual resources. This enables economies of scale and resource utilisation that are efficient, saving users money.

Rapid Elasticity: Resources can be quickly scaled up or down by users in response to changing needs. As a result, purchasing and maintaining physical infrastructure for peak workloads is no longer necessary.

Measured Service: Pay-per-use pricing methods are provided by cloud providers, allowing users to only pay for the services they really utilise. This eliminates the need for up-front capital expenditure and offers cost transparency.